SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

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AI-Based Smart City Surveillance

TECHNICAL PRESENTATION

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Real-time capture from distributed city cameras

- Sources: CCTV, PTZ, thermal and traffic cameras across junctions and public spaces
- Streaming via RTSP / ONVIF; edge devices (Jetson, Raspberry Pi) reduce latency
- Frame sampling and buffering to keep processing in sync with live feeds
- Time-stamped, geo-tagged frames to link events to exact locations
- Secure transport (TLS) and role-based access to protect privacy

Improving frame quality before analysis

- Noise reduction: Gaussian / bilateral filtering for low-light and sensor noise
- Contrast enhancement: histogram equalization and CLAHE for night and fog
- Video stabilization to remove camera shake from wind and vibration
- Resizing, normalization and color-space conversion (BGR to RGB / HSV)
- Region-of-interest masking to ignore irrelevant areas and cut computation

Locating people, vehicles and other objects

- Detectors: YOLO family, SSD and Faster R-CNN trained on COCO / city datasets
- Single-stage models (YOLO) give real-time speed; two-stage give higher accuracy
- Classes: pedestrians, cars, buses, bikes, bags and unattended objects
- Output: bounding boxes, class labels and confidence scores per frame
- Non-maximum suppression removes duplicate detections

From detection to identification

- Multi-object tracking (DeepSORT / ByteTrack) assigns persistent IDs
- Recognition: license plate reading (ANPR) and vehicle type / color classification
- Face or person re-identification only where legally authorized
- Model optimization with TensorRT / ONNX and quantization for edge speed
- Transfer learning and fine-tuning adapt models to local conditions

Separating foreground from background

- Each pixel modeled as a mixture of Gaussian distributions over time
- Pixels that fit dominant Gaussians are background; others are foreground
- Model parameters update continuously to adapt to lighting changes
- OpenCV: MOG2 background subtractor with shadow detection
- Morphological opening / closing and contour filtering clean the foreground mask

Measuring how objects move between frames

- Lucas-Kanade (sparse) tracks key feature points across frames
- Farneback (dense) estimates motion vectors for every pixel
- Flow magnitude gives speed; flow direction gives movement path
- Detects crowd flow, wrong-way vehicles and sudden speed changes
- GMM mask plus optical flow reduces false alarms from moving background

Understanding behavior from visual data

- Features: trajectories, speed, direction, dwell time and object interactions
- Learning normal patterns of each location by time of day
- Action recognition with CNN-LSTM, 3D CNN or video transformers
- Crowd density estimation and flow pattern analysis
- Zone rules: line crossing, restricted-area entry and loitering

Flagging deviations from normal behavior

- Anomaly detection with autoencoders and one-class models trained on normal video
- High reconstruction error or low likelihood marks a frame as abnormal
- Target events: fights, falls, running crowds, abandoned objects, accidents
- Anomaly score thresholds tuned to balance missed events and false alarms
- Temporal smoothing confirms events across several frames

Building the end-to-end pipeline

- cv2.VideoCapture reads live streams; threaded capture avoids frame drops
- Pipeline: preprocess, detect, track, motion analysis, event classification
- cv2.dnn module runs deep models; GPU (CUDA) acceleration raises FPS
- Annotated output: boxes, IDs, trajectories and flow overlays
- Target: 25-30 FPS per stream with multi-camera scaling

Turning detections into action

- Rule engine triggers alerts when an abnormal event passes its threshold
- Notifications via SMS, email, push and control-room dashboard alarms
- Alert packet: event type, camera ID, location, timestamp and video clip
- Event logging in a database for audits, reports and model retraining
- Privacy safeguards: encryption, access control and data retention limits